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# Data Ethics: Big Data, Algorithmic Bias, and Research

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## What is “Big Data”?

“Big Data” is a term which refers to the information collected from large numbers of users or other entities and analyzed collectively. Often, the goal of big data is to predict individual user behavior from patterns developed based on demographically similar users.

- Users’ information is made into a valuable product through “surveillance capitalism.”
- Usage of big data continues to grow, especially by large companies.
- Because this data is often privately controlled, effective oversight can be very difficult or even impossible.

## Questions to Consider

- How are we being represented online?
- Where is data about our lives coming from, and how is it being collected?
- Who is using our data and for what purposes?
- How might our data be used in the future?
- How does “big data” impact our daily lives?

## Online Presence & Data Privacy

Big data impacts many facets of our daily lives. Audio and visual streaming services track user choices to redirect your experiences in real-time (providing “what to watch next” recommendations or endless music mixes). The healthcare industry uses it to keep records of your medical history and track the effectiveness of treatments across demographics. Big data is also used in:

- Shopping and marketing.
- Travel and transportation.
- Education and employment.
- News and information services.
- Public policy and safety.

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Platforms which use big data often collect a greater amount of data than users typically realize. This data continues to become more granular (more specific and intimate) as services increase their usage of big data. For example, TikTok's user agreement allows it to collect biometric data such as faceprints and voiceprints to use for content and ad recommendations.

- Learn about your Google location history from this [resource on location tracking published by Google](#).
- Developed by *The Markup*, the [real-time privacy inspector called Blacklight](#) allows you to reveal tracking technologies hidden in websites.
- Facebook, Google, Instagram and TikTok all have features which allow you to download a copy of your personal data.

### Ethics and Algorithmic Bias

Algorithms (a set of processes or rules which a computer follows) are used to collect, parse, and analyze big data. These algorithms are not neutral, but are the result of programmatic choices. Categories used to collect data often render certain persons, identities, qualities, or characteristics invisible.

- They can reinforce systemic, political, and/or cultural biases.
- When this big data becomes the basis of policy decisions or resource allocation, the consequences can be significant.
- When data reflects biased realities, algorithms will continue to reproduce trained outcomes.

When training algorithms, we must ask what kinds of data are being counted, because “what gets counted counts.”<sup>1</sup>

### Scholarly Biases and Archival Silences

How can we challenge biases, assumptions, and norms within our academic research? Consider:

- Whose voices and expertise are valued and heard?
- What kinds of data are prioritized in scholarship, and how/how often are they used?
- Whose voices and experiences and bodies can we easily find in the historical record, and whose are missing?
- What other sources of information might help fill in gaps in the ‘official’ records found in archives and academic discourse?

When considering archives and the “historical record,” ask yourself:

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<sup>1</sup> Joni Seager quoted in D'Ignazio, C. and L. Klein. (2020) [Data Feminism](#). Cambridge: MIT Press.

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- What information gets saved, and what doesn't?
- Who makes the decisions about what can and cannot be included in "official" records?
- Whose voices, bodies, and experiences are missing from the historical record?
- How can our work be a response to, or disruption of these silences?

### Data Ethics Moving Forward

How are data ethics relevant for you?

- You will likely be collecting, analyzing, and using data in some way.
- Consider from where, and by what methods your data was collected (were people aware of the process, what biases or choices were made in the collection process, who benefits or might be harmed by its collection?).
- Critically examine studies which rely on big data.
- Be mindful of infographics and data visualizations (what is obvious or obscured?).

### Artificial Intelligence (AI) Considerations

Big data and AI are together revolutionizing the scale and speed for mining, analyzing, and monetizing users' information, in areas like marketing and finance and even healthcare. If you are going to use AI—or use a site that has built-in AI features—consider how it derives information, what/whom its sources are, what information it is privileging and/or leaving out, and how your AI usage is contributing to big data. In addition, consider which resources are depleted and communities harmed in the development and use of AI models. The Union of Concerned Scientists' [review of AI](#) and related research on AI's [environmental impacts](#) show that large data centers require enormous amounts of energy—typically combusted from dangerous fossil fuels—and billions of liters of freshwater to keep data processors from overheating. Often, large data centers are located in areas where resources are already scarce, and can result in increased competition for local water, higher energy costs for public ratepayers, higher housing costs, and the displacement of historically-marginalized communities (see the [Climate Justice Alliance](#) for more information). For more perspectives on the relationship between AI and big data, see: this webpage explaining in more detail [the connection between big data and AI](#); the section "[biased and inaccurate output](#)" on UCLA's page for AI in teaching and learning; and the [AI Incident Database website](#), which tracks unethical uses of AI in real time. For more information on Northeastern University AI policies as well as AI tutorials and tips, visit the [Northeastern Center for Advancing Teaching and Learning Through Research](#).

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The influence of AI and the range of its capabilities is rapidly changing, so please be aware that any considerations listed here should be taken in tandem with your current knowledge of these subjects. Please also note that you should follow your professor's stated guidelines for AI use in your class.